

Assignment_GLS_1.0

1. Synthesis

- Refer to Lab_GLS_1.0. Set the generate variable to 0 in *alu8_top.v* to instantiate the **Adder** module.
- Generates key files for GLS:
 - *alu8_top_netlist.v* (*Gate-level netlist*)
 - *alu8_top.sdc* (*Design constraints*)
 - *alu8_top.sdf* (*Timing and delay information*)

2. Gate-Level Simulation (GLS)

- **TB Setup:** Comment out multiplier testcases, uncomment **Adder** testcases, and dump *Adder.vcd*.
- **Required Files:**
 - *Gate-level netlist* (*alu8_top_netlist.v*),
 - *alu8_top.sdf*,
 - *Testbench* , and
 - *standard cell model* (*Lib.v*).
- **Run Command (QuestaSim):**

vsim -do run.do &

3. Power Analysis

- Run PrimeTime power analysis using the dumped *Adder.vcd* to estimate dynamic power consumption.
- **Required Files:**
 - *Adder.vcd*,
 - *alu8_top_netlist.v*,
 - *alu8_top.sdf*,
 - *alu8_top.sdc*, and *Lib*
- **Run Command (PrimeTime):**

pt_shell -f PT.tcl | tee pw.log

4. Deliverables:

- **Synthesis results:**
 - *alu8_top_Netlist.v*
 - *alu8_top_Netlist.ddc*
 - *alu8_top.sdc*
 - *alu8_top.sdf*
- **Synthesis reports:**
 - *Setup.rpt*
 - *Hold.rpt*
 - *Constraints.rpt*
 - *Area.rpt*
 - *Power.rpt*
- **PDF file**
 - *Demonstrates successful GLS execution, including the simulation waveform and results log.*
- **Power report:**
 - *Adder_Power.rpt*